

Assignment 1

Group Number 11

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1 Part A: Research Proposal

1.1 Context

As embodied AI develops, robots are increasingly expected not only to execute predefined commands but also to acquire new skills for physical tasks. This trend was clearly reflected at the 2026 World Artificial Intelligence Conference (WAIC 2026), where more than 300 embodied-intelligence robots demonstrated capabilities such as object manipulation, assembly, and collaboration [1]. This raises a practical question: when a robot needs to learn a new task, what is the most effective way to communicate the task to it?

One common approach is to provide instructions in language. However, describing a physical task in text may not fully capture the sequence, timing, and execution of actions. Learning from Demonstration (LfD) offers an alternative in which robots learn task knowledge from examples of human behaviour [2, 3]. A human demonstration can provide information about how a task is actually performed, rather than only describing the desired outcome.

However, the availability of additional information does not necessarily mean that a robot will perform a task more efficiently. It is therefore interesting to test whether adding a human demonstration to a text instruction actually improves task performance compared with text instructions alone. A randomized experiment can provide a causal test of this effect by keeping the task, robot, and text instruction constant while varying only whether a demonstration is provided.

1.2 Research Question

Does adding a human demonstration to a text instruction reduce the time an embodied AI robot needs to complete a manipulation task?

1.3 Hypotheses

Alternative hypothesis (H_1): Robots that receive a text instruction together with a human demonstration will complete the manipulation task faster than robots that receive the text instruction alone.

Null hypothesis (H_0): Adding a human demonstration to a text instruction has no effect on the robot's task completion time.

1.4 Study Design

We propose a simulated randomized controlled experiment with 100 independent robot trials. The trials will be randomly assigned to two groups of equal size. In the text-only condition ($n = 50$), the robot receives a standardized textual instruction describing how to perform the manipulation task. In the text + demonstration condition ($n = 50$), the robot receives the same textual instruction together with a recorded human demonstration of the task. The robot model, hardware, task, environment, and text instruction will be identical across conditions. The presence of the human demonstration is therefore the only manipulated factor.

1.5 Variables and Sample

The independent variable is **instruction condition**, a nominal categorical variable with two possible values: text-only and text + demonstration. The dependent variable is **task completion time**, measured in seconds. It is a continuous ratio-level variable with a theoretical minimum of zero seconds and positive observed values. The total sample size is $N = 100$, with 50 trials per condition.

1.6 Analysis Pipeline

First, we will calculate the mean and standard deviation of task completion time for each condition and inspect the distributions for extreme values. We will then compare the two groups using an independent-samples t -test with instruction condition as the grouping variable and task completion time as the outcome. The significance level will be set to $\alpha = 0.05$. We will report the mean difference between conditions and the corresponding p -value. A statistically significant negative mean difference, with faster completion in the demonstration condition, will be interpreted as evidence supporting H_1 .

References

- [1] World Artificial Intelligence Conference, “World artificial intelligence conference 2026,” 2026, shanghai, China, July 17–20, 2026.
- [2] B. D. Argall, S. Chernova, M. Veloso, and B. Browning, “A survey of robot learning from demonstration,” *Robotics and Autonomous Systems*, vol. 57, no. 5, pp. 469–483, 2009.
- [3] H. Ravichandar, A. S. Polydoros, S. Chernova, and A. Billard, “Recent advances in robot learning from demonstration,” *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 3, pp. 297–330, 2020.